Thermal enclosure and a slow sensor

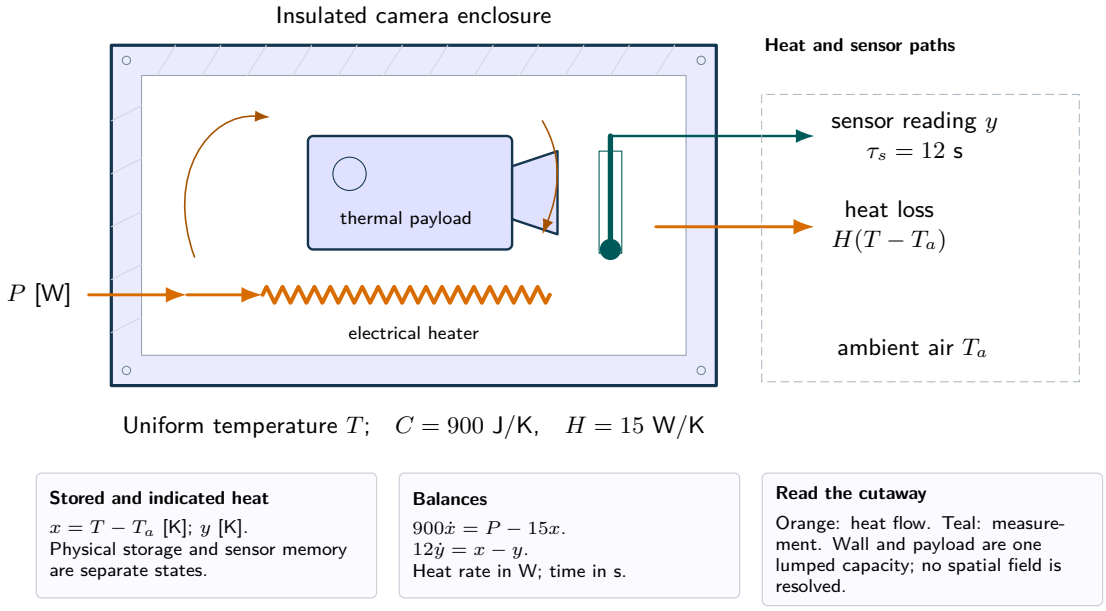
Original learning collection – November 2026

Evidence note. Original teaching scenario. Numerical parameters are deliberately chosen for learning. All generated data and curves are synthetic; no physical measurement is asserted.

1 Design brief

A camera enclosure receives a 40 W heat increment. Use thermal capacitance $C_{\mathrm{th}} = 900\,\mathrm{J/K}$, conductance $H = 15\,\mathrm{W/K}$ and sensor time constant $\tau_s = 12\,\mathrm{s}$. Starting at ambient temperature, derive physical rise $x = T - T_a$ and sensor indication y, their poles and final values, and the sensor error after one enclosure time constant.

2 Physical system



Cutaway of a lumped thermal enclosure. Orange arrows carry heat; the probe measures a lagged temperature. The drawing fixes the energy boundary, not a real device geometry.

3 Energy balance

Let $x = T - T_a$ denote temperature rise in kelvin and P the incremental heat input in watts. With constant ambient temperature and a spatially uniform enclosure,

$$C_{th}\dot{x} = P - Hx, \qquad \tau_s\dot{y} = x - y.$$

The enclosure time constant is $\tau = C_{th}/H = 60$ s and its static gain is $1/H = 1/15$ K/W. The sensor does not change the enclosure energy balance.

4 Cascade model

Taking Laplace transforms with zero initial deviations gives

$$G_x(s) = \frac{1/15}{60s + 1}, \qquad G_y(s) = \frac{1/15}{(60s + 1)(12s + 1)}.$$

The physical and measured responses share the final rise $40/15 = 8/3$ K, but the measured path adds the pole $-1/12$ to the thermal pole $-1/60$.

5 Closed-form step responses

Integrating the energy balance gives $x(t) = \frac{8}{3}(1 - e^{-t/60})$. Partial fractions of $G_y(s)40/s$ give

$$y(t) = \frac{8}{3}\left[1 - \frac{60e^{-t/60} - 12e^{-t/12}}{48}\right].$$

Both start at zero; $\dot{x}(0) = 40/900$ K/s while $\dot{y}(0) = 0$. Evaluate these expressions at $t = 60$ s rather than treating 63.2 percent of the measured final value as the enclosure time constant.

6 Model boundaries

Heat leakage is linearized and the enclosure is assumed well mixed. Radiation, internal temperature gradients, sensor self-heating and changing ambient temperature are omitted. Every curve is a simulation with invented design parameters.

7 What to learn from the experiment

A slow sensor can hide a thermal excursion even when its final calibration is exact. Compare physical and measured states before estimating a time constant.

8 System sketch

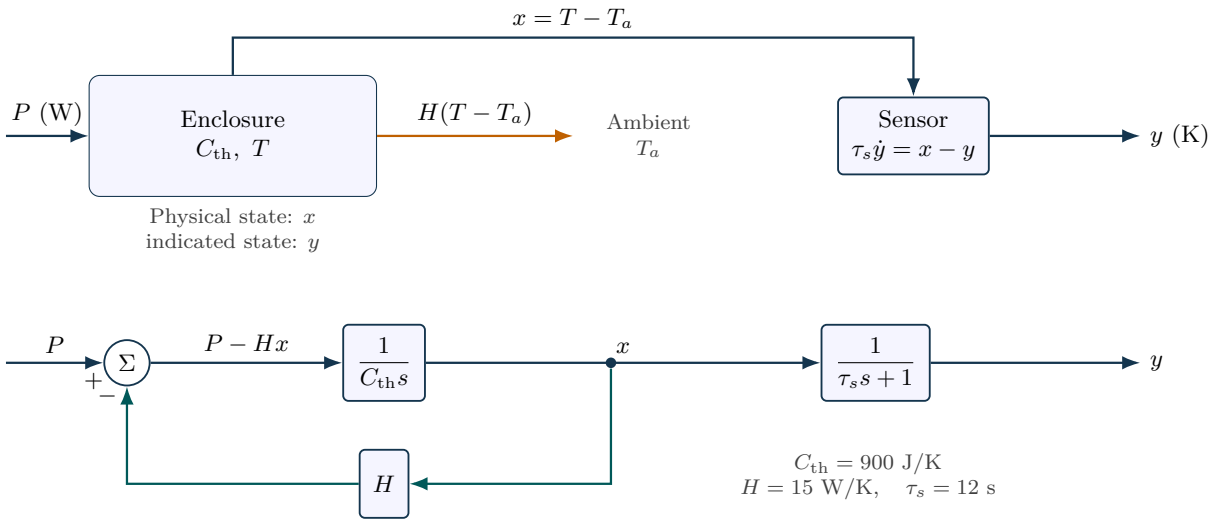
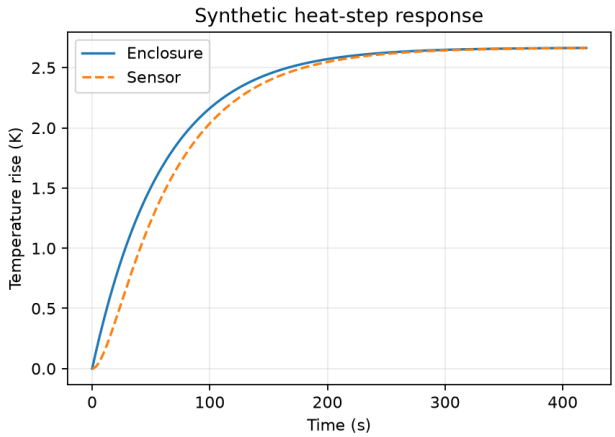
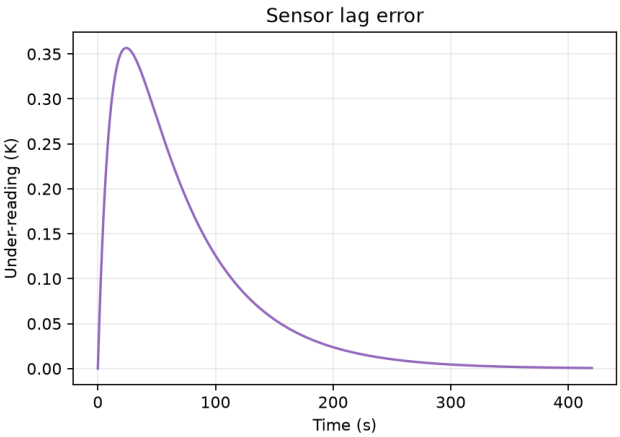


Figure 1. Thermal storage, heat leakage and sensor dynamics are separate elements. The lower loop implements the energy balance, not an added temperature controller.

9 Executed computational evidence

```
Thermal time constant: 60.000 s; final rise: 2.666667 K
Sensor under-reading at 60 s: 0.240761 K
```





10 Method references

The equations use standard balance laws and feedback-control methods. Further explanation: https://apmonitor.com/pdc/ and https://fbsbook.org/. Implementation documentation: https://python-control.readthedocs.io/en/0.10.2/. See the repository reference catalog for individual topics and videos.